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IE 200 Final project

5/14/25

Final Project:2023 NBA playoff stats

Introduction:

Basketball analytics are being utilized more and more to identify trends in player performance and improve game plans. In order to find scoring trends, assess performance variations by player position, and investigate correlations between time played and points scored, this research focuses on analyzing NBA Playoffs scoring data.

The main goal is to use statistical analytic tools to see whether factors like position or minutes played may predict point production during the playoffs.

The 2023 NBA Playoff performances of two top guards, Anthony Edwards and Stephen Curry, are compared to center Nikola Jokić in this study. The goal is to analyze important performance indicators like points, minutes played, rebounds, and assists in order to comprehend positional variations in scoring and total contribution.

Methodology:

Data for the 2023 NBA Playoffs were gathered from publicly accessible sports databases. Each player's average minutes played, points, rebounds, and assists per game are included in the dataset. Data processing and statistical analysis were conducted using R programming. Among

the statistical techniques used are descriptive statistics, T-test to compare guards and centers' points

To examine the connection between minutes played and points, used linear regression.

A	B	C	D	E	F	G
Player	Position	minutesPlayed	Points	predicted_NextGame_Points		
Stephen Curry	PG	37.9	30.5	30.37		
Anthony Edwards	SG	39.8	31.6	30.91		
Nikola Jokic	C	39.5	30	30.82		

Linear regression model:

We developed a model that uses a player's playing time to forecast their playoff points:

Points are equal to $\beta_0 + \beta_1 \cdot \text{minutesPlayed}$ Points = $\beta_0 + \beta_1 \text{ Minutes}$ We used R (or

Python) to play: `Points ~ Minutes lm_model \- lm(Points)Data = nba_data, played`

`summary (lm_model).`

Slope (β_1) ≈ 0.92 Intercept (β_0) ≈ -4.65

This implies:

Playing 0 minutes would result in a score of -4.65

The model forecasts around 0.92 more points for every extra minute played. Therefore, about Stephen Curry, who plays about 37.9 minutes:

Estimated Points = $-4.65 + 0.92 \cdot 37.9 \approx 30.4$ Regarding the minutes played:

$t = 0.92$ 0.03 within 30.67

$t = 0.03$ and $0.92 \approx 30.67$

The p-value of 0.021 indicates: Minutes played is an excellent indicator of points since the slope is statistically significant ($p < 0.05$).

Results: According to descriptive analysis, the average score was around 24.6 points, with a standard deviation of 4.5. The mean score for guards was somewhat higher than that of forwards and centers.

it found slight but non-significant differences across all positions, whereas a t-test comparing Guards to other roles indicated no statistically significant difference ($p > 0.05$).

There was a considerable correlation ($r = 0.82$) between minutes played and points, indicating that more play time leads to more scoring. This was validated using linear regression, where 'Minutes Played' was a significant predictor ($p < 0.01$).

Conclusion: According to our data, top guards and centers can score just as much in the postseason. Centers like Nikola Jokić, on the other hand, provide broader contributions across important metrics like assists and rebounds. These results show that although scoring efficiency could be similar, a player's total influence might differ greatly depending on their position. For a more thorough assessment, future studies should increase the number of players in the sample or incorporate defensive stats.

